A simplified approach to model grid-forming controlled MMCs in power system stability studies

Juan Carlos Gonzalez-Torres, Rayane Mourouvin, Kosei Shinoda, Ahmed Zama, Abdelkrim Benchaib
Supergrid Architecture & Systems
SuperGrid Institute SAS
Villeurbanne, France
juan-carlos.gonzalez@supergrid-institute.com

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***Abstract*—This paper validates a phasor model of a grid-forming controlled Modular Multilevel Converter (MMC) suitable for large AC grid dynamic studies. The adopted MMC model represent with accuracy the energetic exchanges between the AC side, the DC side and the equivalent internal energy inside the MMC (represented by a single state variable). After description of the converter model, the adopted control strategy is presented and discussed. The simplified model of the converter with its control is validated against a detailed electromagnetic transient model in different control modes (active power control and DC voltage control). Finally, the compatibility of the simplified model with large scale AC dynamic simulations under phasor approximations is validated through the simulation of the Nordic 44-bus test system including grid-forming MMCs.**

***Index Terms*—Power systems, HVDC, Grid-forming, MMC.**

I. Introduction

To meet the objectives of decarbonisation of the electricity sector and satisfy the growing demand for energy, the massive integration of renewable energies is essential. These energy sources are often connected to the grid via power electronics-based converters, whose dynamics are different from those of classic generation units, bringing new challenges for system operators. Indeed, under scenarios with a high share of non-synchronous generation, potential issues have been identified regarding the operation and control of the power systems [1].

A solution that is being investigated by industry and academia to operate Voltage Source Converters (VSC) in power systems under this evolving configuration are the so-called *grid-forming* control strategies [2]. Grid-forming control has shown its benefits for allowing more power electronics interfaced sources to be integrated in the grid [3], [4] compared to classical *grid-following* control. Despite its advantages in certain scenarios, the concept of grid-forming converters is still relatively controversial, and there are many techniques to achieve it (Virtual Synchronous Machine [5], droop control [6] among others [7]).

Compared to classical grid-following converters, the dynamics of grid-forming converters have shown a dependency on the electromechanical dynamics of the grid. Most works in the literature propose and tune grid forming controls through a single converter infinite bus approach using the equivalent grid impedance [6], [8], thus neglecting the electromechanical dynamics of the grid. Other works have used an equivalent generator to represent the frequency dynamics of the grid [9], therefore neglecting some possible electro-mechanical oscillations observed at the point of common coupling of the converter. Tools as the one presented in this article are necessary to analyse the response of grid forming converters when connected to complex grids.

At the transmission level, renewable energy sources are likely to be in large centralised utilities (e.g. offshore wind farms) expected to inject high amounts of power to the grid. For these high power applications, the Modular Multilevel Converter (MMC) has proven to be the more adapted topology [10]. Thanks to its modular architecture, when the MMC is properly controlled, the total stored energy inside the sub-modules capacitors can be controlled (known as energy-based control as in [11]), offering an additional degree of freedom.

In recent literature, grid-forming controls have been adapted to take into account the specificities of MMCs [7], [12]. However these grid-forming controls for MMCs have mainly been tested under simplified AC grid models (single converter infinite bus or single converter single machine systems). Therefore, providing a tool where a grid-forming MMC connected to a complex AC grid can be simulated without the computational cost of Electromagnetic Transient (EMT) simulations is the main motivation of this work. To do so, a grid-forming controlled MMC model under phasor approximations adapted for large scale simulations is presented. This model is based on the MMC model presented in [13] and the grid-forming controls discussed in [6], [7]. In this article, the MMC model is implemented using the Modelica language, so it can be tested in combination with the OpenIPSL library [14], [15], developed for dynamic simulation of AC networks.

The subsequent sections are organised as follows. Section II presents the adopted model of the MMC as well as the adopted grid-forming control. In Section III, the proposed model is validated against a detailed EMT model. In section IV, a dynamic simulation of a large grid including MMCs is performed. A discussion on how the provided tools can be exploited further is done in Section V.

II. RMS modelling of the grid forming MMC

A. Modelling the converter under phasor approximations

The adopted simplified MMC model is the one derived in [13], [16]. Different from previous propositions, this model

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allows to maintain the average dynamics of the stored energy in the MMC sub-modules. This is particularly important when the converter is controlled following an energy-based philosophy [11]. To achieve this representation, the MMC is modelled as the association of an AC/DC converter and an equivalent DC/DC interfaced by capacitor as illustrated in Fig. 1. The equivalent capacitor C_{eq} interfacing the AC and DC sides represents the equivalent aggregated sub-module capacitance of the converter.

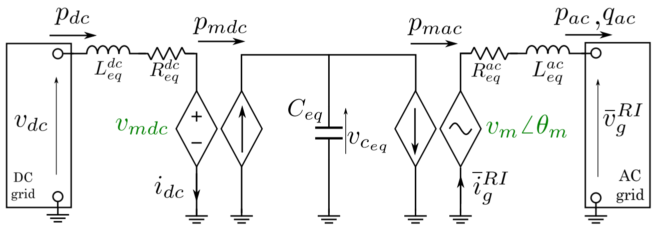

Fig. 1. MMC model under phasor assumptions

On the AC grid, all quantities are represented by time-varying phasors on a common frame for all the AC network noted as *RI frame* in this paper. The AC side of the MMC is then modelled as controllable phasor voltage source $\bar{v}_m^{RI} = v_{mR} + jv_{mI} = v_m \angle \theta_m$ coupled to the PCC (with voltage $\bar{v}_g^{RI} = v_{gR} + jv_{gI}$) through an equivalent resistor R_{eq}^{ac} and inductance L_{eq}^{ac}. Similarly, the converter DC side is represented as a DC voltage source v_{mdc} connected to the DC grid (with voltage v_{dc}) via an equivalent resistance R_{dc}^{eq} and inductance L_{eq}^{dc}. The currents on the DC side and AC side of the converter are respectively noted i_{dc} and $\bar{i}_g^{RI} = i_{gR} + ji_{gI}$. The converter is considered lossless, therefore at each equivalent conversion stage (AC/DC and DC/DC conversion), the active power is conserved (noted p_{mac} and p_{mdc} in the Fig. 1).

The dynamics of simplified model of the converter are therefore described by the following equations:

$$
\begin{aligned}
v_{mR} - v_{gR} &= L_{eq}^{ac}\frac{di_{gR}}{dt} + R_{eq}^{ac} i_{gR} - \omega L_{eq}^{ac} i_{gI} \\
v_{mI} - v_{gI} - &= L_{eq}^{ac}\frac{di_{gI}}{dt} + R_{eq}^{ac} i_{gI} + \omega L_{eq}^{ac} i_{gR} \\
v_{dc} - v_{mdc} &= L_{eq}^{dc}\frac{di_{dc}}{dt} + R_{eq}^{dc} i_{dc} \\
C_{eq}\frac{dv_{c_{eq}}}{dt} = \frac{p_{mdc} - p_{mac}}{v_{c_{eq}}} &= \frac{v_{mdc} i_{dc} - v_{mR} i_{gR} - v_{mI} i_{gI}}{v_{c_{eq}}}
\end{aligned}
\tag{1}
$$

The modulated DC and AC voltages (v_{mdc}, v_{mR}, v_{mI}) are the control inputs of the system. The calculation of C_{eq}, R_{eq}^{ac}, R_{eq}^{ac}, L_{eq}^{ac} and L_{eq}^{dc} as function of physical MMC parameters can be found in Appendix A.

B. Control of the MMC

The structure of the adopted control is depicted in Fig. 2. First, it is observed that this structure allows the converter either to have a power reference p^* given by the operator (P_{ac}-mode) or to control the voltage on the DC side (V_{dc}-mode) via a PI controller that uses p^* as control input.

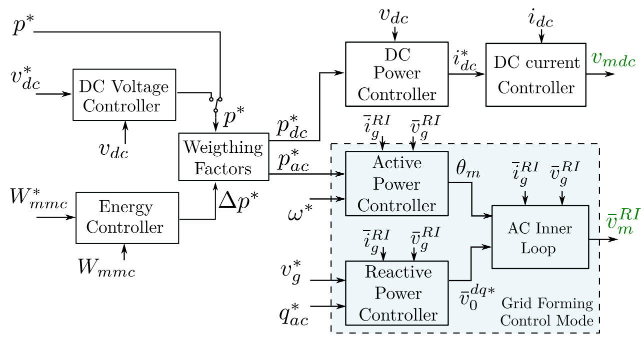

Fig. 2. Global structure of the MMC control

In a parallel stage, the Energy Controller, uses a PI control to maintain the energy inside the MMC within an acceptable level [16]. This energy is controlled through the modulation of the power reference Δp^*. The Weighting Factors block distributes this modulation between the AC and DC power references (p_{dc}^*, p_{ac}^*) as function of a tunable coefficient α as follows:

$$
\begin{aligned}
p_{dc}^* &= p^* + \alpha \Delta p^* \\
p_{ac}^* &= p^* - (1-\alpha)\Delta p^*
\end{aligned}
\tag{2}
$$

The reference p_{dc}^* is sent to the DC power controller, while the reference p_{ac}^* is sent to the control stage responsible of the grid-forming mode. The DC power controller consists in a current reference calculator that computes $i_{dc}^* = p_{dc}^*/v_{dc}$. The DC current controller that generates v_{mdc} is a classical PI controller set to track the current reference [16].

The adopted grid-forming control is based on the ones proposed in [6], [17]. This control can be decomposed in three blocks: the active power controller, the reactive power controller and the inner loop control. The active power controller, also known as power synchronisation loop, [6] is depicted in Fig. 3. This block uses the active power variations to estimate the necessary frequency and angle of the modulated AC voltage that allows to keep the converter in synchronism with the grid.

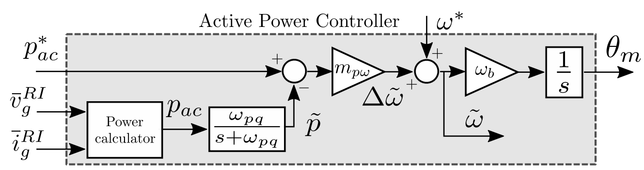

Fig. 3. Grid forming - Outer loop - Active power control

The reactive power controller is used to determine the amplitude of the modulated voltage v_{md}^* via the reactive power variations, following the droop Q-V_{ac} philosophy as in [6].

Classical 2-level VSCs are equipped with an LC (or LCL) filter to enhance their harmonic performance. The use of this filter allows the use of cascaded controllers to control the voltage at the filter capacitor (set to track $v_{md}^* \angle \theta_m$), via the control of the currents at the inductance filter [3]. Since MMCs do not require LC filters, a cascaded structure is not suitable, especially if the AC network is not highly

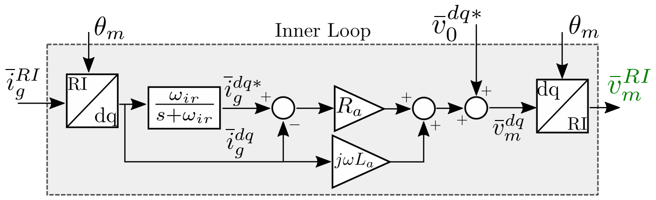

Fig. 4. AC inner loop

capacitive. It is however possible to take advantage of the physical behaviour of the converter as a voltage source to directly impose the generated voltage reference by the grid-forming loops ($v_d^* \angle \theta_m$). This concept is known as AC direct voltage control [7]. When adopting this strategy, an inner loop is however necessary in order to provide some damping to the grid currents [6]. Improvements have been recently proposed to the inner loop, especially regarding over-current limitation [7], [12]. The adopted inner loop is depicted in Fig. 4.

III. Validation of the Simplified Model

In this section, the presented average model of the grid-forming MMC is validated against its equivalent detailed EMT model. The simplified model of the MMC has been implemented using Modelica language and it has been coded to be compatible with the OpenIPSL library dedicated to dynamic simulations of AC networks [15]. The detailed EMT model of the converter corresponds to a 200-level MMC implmented in Matlab/Simulink. A detailed description of the converter and the low level controllers can be found in [18].

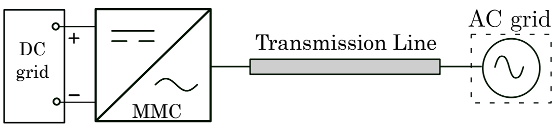

Fig. 5. Single converter infinite bus test system with ideal DC source

The topology of the test system is shown in Fig. 5. It consists in an MMC connected through a transmission line to an ideal AC voltage source. Two control modes are tested: the P_{ac}-mode and the V_{dc}-mode. The configuration of the DC side varies from an ideal source to an equivalent capacitor with an external DC current injection depending on the tested mode. The parameters of the simulation are listed in Appendix B.

For each control mode, three different cases are compared:

- EMT : Detailed EMT model of the test system
- Phasor-SL (Static model of the Line): The simplified model of the MMC is connected to the AC source through a line whose dynamics have been neglected (the line is represented by algebraic equations as done in classical transient stability software).
- Phasor-DL (Dynamic model of the Line): The transmission line dynamics have been included using the dynamic equations of a π-section on the *RI* frame [19]. This kind of models are essential to reproduce the behaviour of the converters when they are connected to weak grids.

A. Active power control mode

The converter is set into P_{ac}-mode and a -0.3 p.u. step is applied to the power reference at $t = 0.1$s followed by a 0.6 p.u. step at $t = 1$s. For this test, the DC side is represented by an ideal voltage source. The active power delivered by the converter to the AC grid, is shown in Fig. 6. It is observed that the response of the simplified models (phasor-SL and phasor-DL) is very similar to the response of the detailed EMT model. The active power variation leads to a temporary imbalance of the MMC internal stored energy. The energy controller acts through the DC current references in order to keep the energy at 1 p.u. The total internal energy and the DC currents are depicted in Fig. 7 and Fig. 8 respectively. For the quantities analysed in this test, the simplified models manage to reproduce the average behaviour of the detailed model.

B. DC voltage control

The converter is set into v_{dc}-mode, and the DC grid is represented by an equivalent capacitor fed by an external current source. A first order filter with time constant of 100ms is used on the external current source as to emulate the behaviour of a power source. Using this setup, a -0.3p.u. and a 0.6p.u. steps of current are injected by the external current source at t=0.1s and t=1s respectively. Fig. 11 shows the voltage of the equivalent DC grid capacitor. When the current step is applied, a power imbalance is created in the DC capacitor, leading to a variation of its voltage. The voltage controller manages to take back the voltage to the reference value by modifying the power

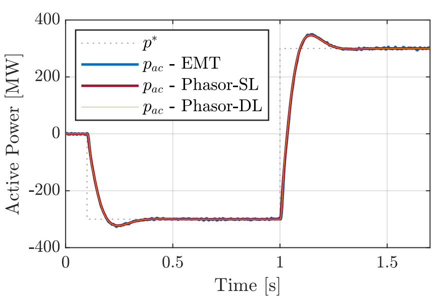

Fig. 6. P_{ac} mode - Active Power

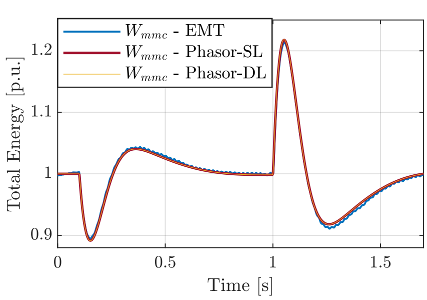

Fig. 7. P_{ac} mode - MMC internal energy

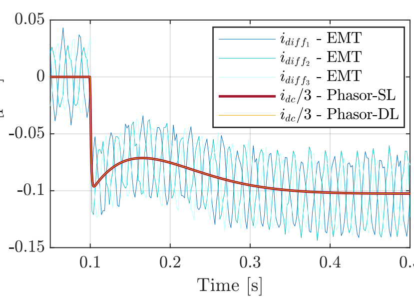

Fig. 8. P_{ac} mode - DC currents

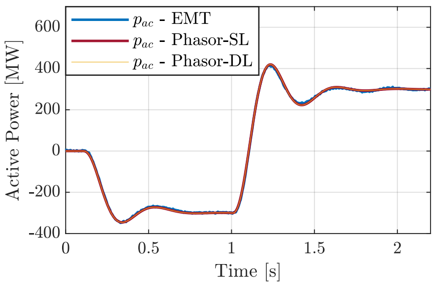

Fig. 9. V_{dc} mode - Active Power

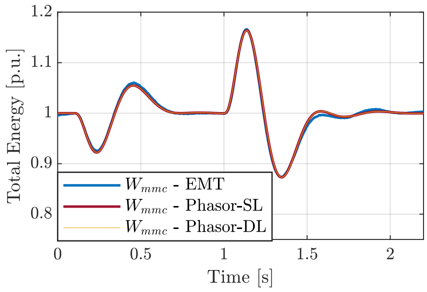

Fig. 10. V_{dc} mode - MMC internal energy

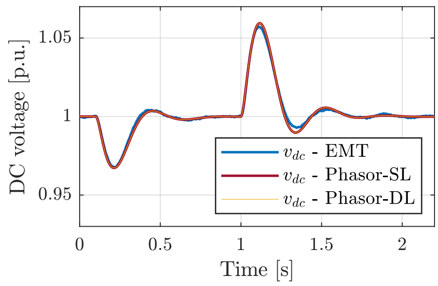

Fig. 11. V_{dc} mode - DC voltage

reference p^* of the converter (Fig. 9). This reference is sent to the AC and DC sides, however since their time response are not the same, an imbalance in the total internal energy is produced (Fig. 10). As the coefficient α in the weighting factors block is set to 1, the energy is then controlled via the DC power. For this test, we observe that the simplified models (Phasor-SL and Phasor-DL) reproduce accurately the energy exchanges of the EMT detailed model.

Compared with classical grid-following schemes, in the analysed grid-forming MMC time responses of the different control loops (see Appendix B) are slower and overshoots are higher. As the active power time response of grid-forming converters is generally slower, it is therefore necessary for the energy and DC voltage loop to be slower. A discussion on the impact of the time responses of the different control loops of the MMC can be found in [20].

IV. Case Study

In this section, the described grid-forming MMC model is implemented in a large AC grid. The test system is a modified version of the Nordic 44-bus system (details can be found in [21]). A diagram of the system is depicted in Fig. 12.

The modifications with respect to the original case are:

- Grid-forming MMCs in V_{dc}-mode are connected to buses 7000 and 5500 through transmission lines ($x_l = 0.166$ p.u. and $r_l = 0.0166$p.u.). On their DC side an equivalent capacitor and a filtered DC current injector are connected. The initial power set-point of these stations is 0 MW. The parameters of the MMC and control are the same as in Section III, with the difference that $\alpha = 0$.
- The generator in bus 3245 is replaced by a grid-forming MMC station in P_{ac}-mode. The active power set-point of this station is the same as the replaced synchronous generator (i.e 205MW). The parameters of the MMC and its control are the same as in Section III
- Transmission lines connecting the MMCs to the nearest generators (depicted in green in Fig. 12) are represented by dynamic π-sections (also known as *hybrid network model* [19]).

The simulated sequence of events is the following :

- At t=2s, 800MW of generation at bus 5300 are tripped.
- At t=2.2s, a DC current injection of 0.1 p.u. on the DC side of the MMCs controlled in DC Voltage Control mode (MMC 7000 and MMC 5300) is activated as to simulate

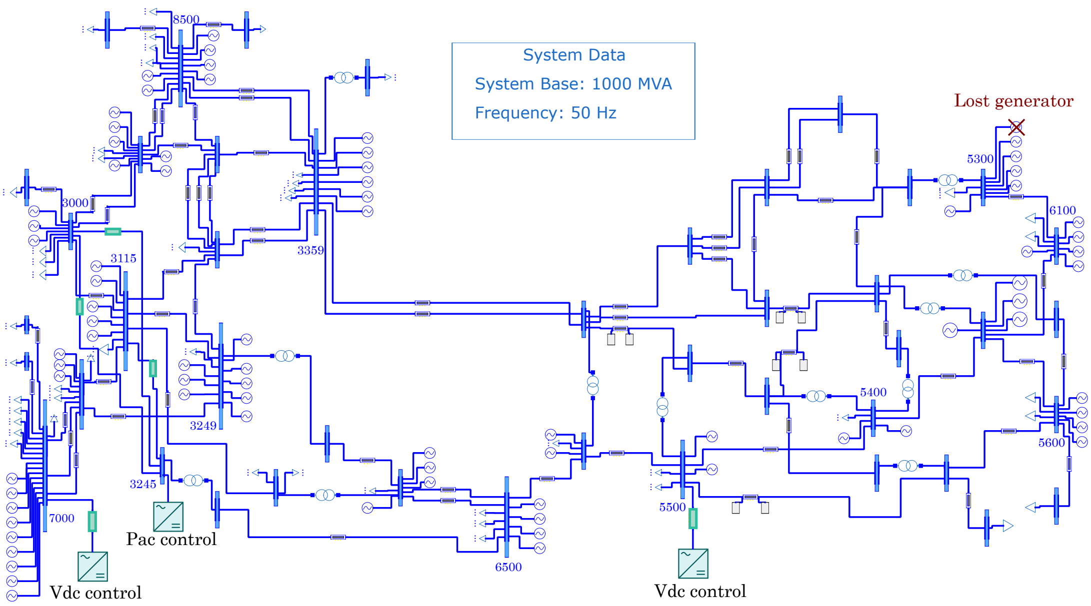

Fig. 12. A modified version of the Nordic 44-bus test system

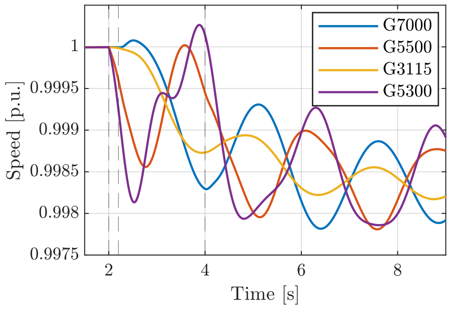

Fig. 13. Speed of some relevant generators

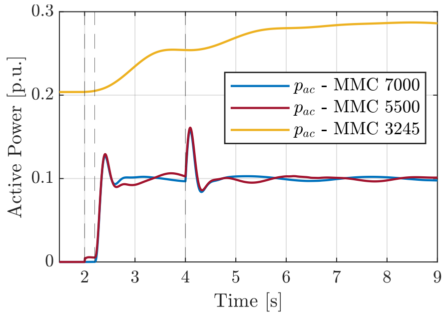

Fig. 14. MMCs active power

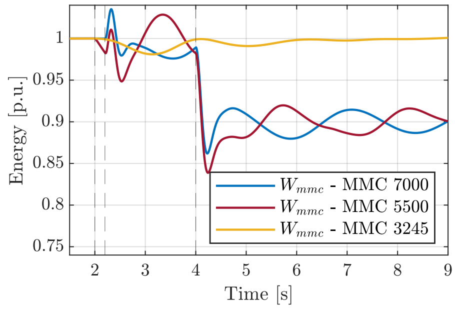

Fig. 15. MMCs internal energy

a triggered fast frequency reserve delivery coming form an external source.
- At t=4s, the internal energy set-point of MMC 7000 and MMC 5300 is set to 0.9 p.u. Indeed, some works propose to discharge a portion of the energy of the MMC on the AC grid following a frequency event [22].

Fig. 13 shows the speed of some generators in the grid. As the global frequency decreases when the disturbance occurs, generators experience electromechanical oscillations around the average frequency. The amplitude of their first swing and RoCoF vary as function of their distance from the event (e.g. G5300 RoCoF is the highest as it is near the lost generator).

In Fig. 14 the active power of the MMCs is depicted. As MMC 3245 is in P_{ac}-mode and the adopted grid-forming control provides a primary response, its active power follows the frequency deviation of the grid. MMC 7000 and MMC 5500, being in V_{dc}-mode, inject to the AC grid the power coming from the external DC current source when the 0.1p.u. DC current step is triggered (t=2.2s). At t=4s, the energy reference W^*_{mmc} of MMC 7000 and MMC 5500 are set to 0.9p.u. As their energy is controlled through the AC power ($\alpha = 0$), this set-point change results in a power pulse delivered to the AC grid. Throughout the event, slight oscillations in the active power of the converters occur as nearby generators keep oscillating around the average frequency.

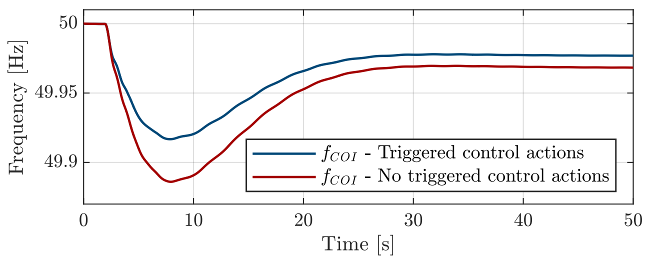

Fig. 16. Comparison of the COI frequency

Fig. 15 shows the internal energy of the converters. As the energy of the MMC 7000 and MMC 5500 is controlled through the AC power and the grid forming control uses the active power modulations to synchronise with the grid, strong interactions are observed. In other words, oscillations in the grid are reflected in the total energy.

Fig. 16 shows in blue the evolution of the frequency of the Centre of Inertia (f_{COI}) for the simulated sequence of events. The red curve represents the frequency in the case where MMC 7000 and MMC 5500 do not deliver triggered frequency containment reserves after the generation trip.

V. Discussion on the Use of the Tool

In the previous section, the simulation package (MMC model + OpenIPSL) was used to assess frequency stability, and to observe the behaviour of the converters facing electromechanical oscillations. The presented package, can be further used to assess the transient stability of the system in the presences of converters, either by assessing the synchronizing capability of the converter [8], or to analyse the impacts of power electronics devices on the transient stability of generators [23] (for these studies, over-current limitation algorithms must be included). Moreover, Modelica-based modelling, offers the possibility to obtain a linear model of the simulated system using embedded tools in Modelica simulators [24]. Small-signal stability studies can be then used in the design of grid-forming controllers taking into account the grids dynamics, or in the design of supplementary controllers as in [25], [26].

VI. Conclusion

A simplified model of an MMC with grid-forming control, adapted for phasor simulations of power system has been presented. A comparison between the adopted model and a detailed EMT model showed the accuracy of the simplified model to represent the energy exchanges between the AC grid, the DC grid and the internal MMC energy. The model has shown to be adapted for dynamic simulations of AC grids under phasor approximations. This has been achieved through the simulation of a modified version of the Nordic 44-bus test system in which the simplified grid-forming MMC model has been included.

Acknowledgment

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Appendix

A. Equivalent elements of the simplified model

The parameters of the simplified MMC model are given by:

$$C_{eq} = 6\,C_{sub}/N_{arm},$$
$$R_{ac}^{eq} = R_f + R_{arm}/2, \qquad L_{ac}^{eq} = L_f + L_{arm}/2,$$
$$R_{dc}^{eq} = 2R_{arm}/3, \qquad L_{dc}^{eq} = 2L_{arm}/3,$$

where N_{arm} is the number of the sub-modules per arm, C_{sub} is the capacitance of a sub-module, L_f and R_f are the transformer inductance and resistance, L_{arm} and R_{amr} are respectively the arm inductance and arm resistance.

B. Parameters of the test system

TABLE I
PHYSICAL PARAMETERS.

Quantity	Notation	Value
Base frequency	ω_b	50 Hz
Reference frequency	ω^*	1 p.u.
AC line-to-line rated voltage	V_b^{ac}	320 kV
DC grid rated voltage	V_b^{dc}	640 kV
DC-side capacitor	C_{dc}	195 µF
MMC rated power	MVA_b	1000 MVA
MMC submodule capacitor	C_{sub}	10 mF
# of submodules per arm	N_{arm}	200
Arm inductance	L_{arm}	0.0758 p.u.
Arm resistance	R_{arm}	0.0078 p.u.
Transformer inductance	L_f	0.18 p.u.
Transformer resistance	R_f	0.001 p.u.
Transmission line reactance	X_L	0.167 p.u.
Transmission line resistance	R_L	0.167 p.u.

TABLE II
CONTROL PARAMETERS.

Quantity	Notation	Value
Active power droop gain	$m_{p\omega}$	0.02 p.u.
Reactive power droop gain	m_{qv}	0 p.u.
Outer loop filter cut-off frequency	ω_{pq}	31.4 rad/s
Inner loop filter cut-off frequency	ω_{ir}	314 rad/s
Damping resistance	R_a	0.2 p.u.
Inductance compensation	L_a	0 p.u.
Energy control response time	τ_{rwh}	0.4 s
Energy reference	W^*_{mmc}	1 p.u.
DC voltage reference	v^*_{dc}	1 p.u.
Energy distribution coefficient	α	1
DC-voltage control response time	τ_{rdc}	0.15 s
DC-current control response time	τ_{idc}	0.003 s
AC-current control response time	τ_{iac}	0.005 s
Damping ratio of all control loops	ζ	0.707

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